

or cable lines that give us, realistically, about a maximum of 10 Kbps download speeds. While service providers abroad do offer connectivity at much higher speed, optical fibre to the home is the Holy Grail Internet users aspire for.

When optical fibre was first introduced, it seemed to be too expensive for use in homes. However, the demand for voice/data/video bandwidth has grown along with developments in laser technology, which will accelerate the progress towards the day when optical fibre becomes a cost-effective as well as high-speed transport medium for home users. The increased use of optical fibre is mostly due to advances in production methods, which has reduced the price while increasing availability of optical elements.

Strands of optical fibre will probably be the only backbone connection for neighbourhood gateways and will carry all voice and data traffic directly into and out of our homes. With the increasing complexity of existing voice, data and video communications infrastructure, it seems practical to merge all the traffic into a single converged network that will work for everything. Optical fibre networks, which has the potential to carry up to 100 terabits per second, is the ideal solution in this situation.

While the use of optical fibre is currently limited to corporate users and some private networks, eventually consumers will be able to get direct connections to high-speed networks.

Technologies such as DWDM (Dense Wave Division Multiplexing) together with new developments in laser technology such as NECSEL (Novalux Extended Cavity Surface Emitting Laser) are going to further increase the capacity of the Internet backbones by making more efficient use of existing optical network technology. Network congestion, a problem currently faced

by ISPs all over the world at peak usage times, would also be alleviated to some extent by the presence of multi-gigabit-bandwidth optical fibre backbones between ISPs, intra-country backbones and inter-country backbones.

Video @ Home

The same single cable into your home that provides you with blazing fast Internet access will also change the way you watch television. Video-on-Demand (VOD) systems provide access to a large quantity of video information over computer networks. Viewers can submit a request to the VOD system over the Internet and view audio and video streams in the comfort of their homes. The audio-video content is provided by streaming the data from a media file server to your computer. This would run off a distributed storage system that has a capacity of many terabytes of digital material. A Digital Asset Management (DAM) system forms a major part of any VOD network and allows the organisation and retrieval of a variety of different data types.

Technologies such as multicasting will power VOD and HTDV-quality live broadcast, which will also be delivered to you off the Internet. This involves using networks in a smarter manner, thus making higher performance levels possible. Here, instead of one server sending multiple streams of the same data to locations on the Internet, a single stream of data is replicated to different locations. This can help relieve congestion considerably at the server end, especially for bandwidth-hungry applications such as streaming video.

Greater reliability

One problem that is faced on existing networks is reliability or quality of service (QoS).

Browsers of the future

Browsers have seen a lot of change as the Internet has developed—starting from text browsers such as Lynx to the first 'graphical' browsers, which were the Mozilla generation, followed by the current breed including IE 6 and Netscape v6.1.

Wondering what is coming next? Here are some examples that provide a peek into tomorrow's browsers.

HyperSPC: It won a 'Concept Browser' competition a couple of years ago.

(www.crd.rca.ac.uk)

ViOS: The Visual Internet Operating System renders 3D environments on the fly. You can walk through this 3D world like in a computer game. (www.vios.com)

CubicEye: This is another 3D browser that organises Web pages in the form of a cube. Users can view the multiple screens that form the walls and the bottom of a box from the inside. (www.cubiceye.com)



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